OFER SHAHAL

Data Product Lead | Project Management | GIS | Data Science

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SUMMARY

Product and technology leader with hands-on experience managing and delivering national-scale geospatial data systems and products — owning the full lifecycle from requirements discovery and system architecture through cross-functional delivery and adoption, coordinating engineering, GIS, and government stakeholders across multiple countries. Strong foundation in structured data, metadata, and information architecture, with a track record of translating complex operational needs into scalable, data-driven products and measurable outcomes. Self-taught in applied AI and LLMs, with hands-on experience building and experimenting with grounded, tool-calling agents on real projects — now looking to grow this into a core product focus, bringing the same structured, requirements-driven approach to AI-enabled products.

SKILLS

Data Science & AI: Data Modelling, Pandas, scikit-learn, Random Forest, XGBoost, SHAP, Optuna, LangChain, Hallucination Mitigation, Tool-Calling, Evaluation, ReAct Agents, LangGraph, Claude, Cursor

Database & Data Engineering: PostgreSQL, PostGIS, DuckDB, Excel, ETL, ELT, DBT

Product & Delivery: Product Requirements, Prototyping & POCs, Requirements Translation, Stakeholder & Cross-Team Coordination, Agile Methodology

Measurement & Dashboards: KPI Definition, Analytics Tools, Dashboard Development (Streamlit, Plotly, Grafana), Data-Driven Decision Support

Programming & Data: Python, SQL, FastAPI

Professional: Analytical Thinking, Curiosity, Self-Learner, Critical Thinking, Cross-Team Communication (Content & Tech), Detail-Oriented

EXPERIENCE

Project & Product Manager — Data & Geospatial Systems
2025 - Present

Sivan Design D.S LTD, Israel / Zanzibar / Nigeria

- Leading the end-to-end project and product for national-scale geospatial data systems (ZanSDI — Zanzibar, Ogun State SDI — Nigeria), owning delivery from requirements through architecture, deployment, and adoption.
- Translating government and operational requirements into data structures, analytical workflows, and system requirements for national geospatial data programs.
- Leading cross-functional technical delivery across developers, GIS specialists, and client stakeholders, aligning data architecture and platform capabilities with business and operational needs.
- Designing data schemas, metadata structures, and system architecture for these SDI programs, including deployment of an open-source geospatial stack (GeoNode, GeoServer, PostGIS).
- Preparing technical documentation, including data model design and integration specifications for multi-machine environments.

Transportation & Geospatial Data Analyst
2022 - 2025

Tel Aviv-Yafo Municipality, Tel Aviv-Yafo

- Structured and automated data pipelines (GTFS, SIRI, e-scooter datasets) to support transportation analytics and decision-support products.
- Applied statistical and spatial analysis to validate findings and translate them into dashboards and planning recommendations.

Geospatial Analyst & Researcher
2023

Freelance — commissioned by Tel Aviv-Yafo Municipality

Analyzed ~1M shared e-scooter trips using statistical modeling and spatial analysis to inform micro-mobility parking policy.

Geospatial Data Analyst
2021 - 2022

Competition Authority, Jerusalem

Applied geographic principles and GIS methodologies in departmental research to enhance decision-making.

GIS Operator
2020 - 2021

Ministry of Environmental Protection, Jerusalem

Managed and enhanced GIS data for the Environmental Planning Dept.

EDUCATION

Certified Data Scientist with Expertise in Generative AI and AI Agents 2025 - 2026
Bar-Ilan University

Master's Degree in Geography 2022 - 2024
Tel-Aviv University
Thesis in Geosimulation lab.

Bachelor's Degree in Geography & Middle Eastern Studies 2018 - 2022
Tel-Aviv University

PROJECTS

Cadaster AI Assistant — Grounded LLM Interface to Structured Data
Researched how an LLM agent should read and rely on structured information sources, and identified a recurring gap between human-readable data and machine-consumable structure. Built a proof-of-concept conversational interface where the agent calls predefined tools over curated dbt models rather than querying freely, directly addressing grounding and hallucination risk. Explored trade-offs between AI accessibility and data governance (Python, FastAPI, LangChain, LangGraph, dbt, PostGIS). [Personal Project, 2026]

Israel Transit AI Agent
Built a ReAct-style LLM agent answering natural-language questions via tool-calling over a live, structured GTFS/SIRI feed — a hands-on exploration of retrieval and tool-use patterns for grounding an agent in real data (LLM Agents, Tool-Calling, DuckDB, Streamlit). [Bar-Ilan Data Science Course, 2026]

Israel Train Stations Dashboard
Designed and built a user-facing interactive dashboard exploring national rail passenger patterns from open government transport data (Streamlit, Plotly, Folium). [Bar-Ilan Data Science Course, 2025]

AWARDS & HONORS

M.A. Excellence Scholarship, The Exact Science Faculty 2024
Tel Aviv University

B.A. Excellence Track Scholarship 2019 - 2022
Tel Aviv University

The Exact Science Faculty Dean's List 2019 - 2020
Tel Aviv University

Best Papers Section 2023
18th Conference on Location Based Services, Ghent, Belgium
Conference presentation “Shared Dockless E-Scooters in the City: System Analysis and the Novel Management Policy” selected for the Best Papers Section.